Set 27: Patient Health Risk Analysis

AIM

To analyze patient health indicators using R programming by creating a scatterplot matrix, Q-Q plot, ECDF plot, and bar chart to examine relationships between health indicators and identify potential health risks.

ALGORITHM

1. Create a dataset containing Patient ID, Age, BMI, Blood Pressure (BP), and Cholesterol.
2. Load the ggplot2 and GGally libraries.
3. Create a scatterplot matrix to examine pairwise relationships between Age, BMI, BP, and Cholesterol.
4. Generate a Q-Q plot to assess whether Cholesterol values follow a normal distribution.
5. Create an empirical cumulative distribution function (ECDF) plot for Cholesterol.
6. Calculate the average value of each health indicator and display the averages in a bar chart.
7. Interpret the relationships and identify patients who may require further clinical assessment.

R PROGRAM

```
patient <- data.frame(
  ID = c("P1","P2","P3","P4","P5"),
  Age = c(25,40,55,35,60),
  BMI = c(22,28,30,26,32),
  BP = c(120,135,145,130,150),
  Cholesterol = c(180,210,240,200,260)
)


pairs(patient[, 2:5], main = "Patient Health Analysis")


qqnorm(patient$Cholesterol, main = "Q-Q Plot of Cholesterol")
qqline(patient$Cholesterol, col = "red")

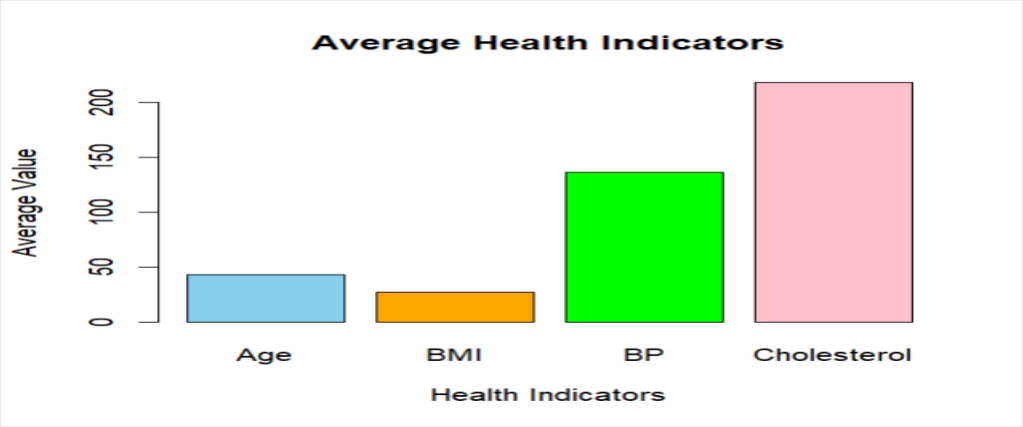

plot(ecdf(patient$Cholesterol),
```

```
    main = "ECDF of Cholesterol",
    xlab = "Cholesterol", ylab = "Cumulative Probability")


avg <- colMeans(patient[, 2:5])


barplot(avg, col = c("skyblue", "orange", "green", "pink"),
        main = "Average Health Indicators",
        xlab = "Health Indicators", ylab = "Average Value")
print(avg)
```

OUTPUT:



RESULT

The scatterplot matrix helps identify relationships among Age, BMI, Blood Pressure, and Cholesterol. The Q-Q plot assesses how closely cholesterol values follow a normal distribution, while the ECDF displays the cumulative distribution.

Average health indicators:

Indicator	**Average**
Age	43
BMI	27.6
Blood Pressure	136
Cholesterol	218

The dataset shows that P5 has the highest age, BMI, blood pressure, and cholesterol values. P5 may warrant further assessment by a healthcare professional. These observations alone do not establish a diagnosis or determine an individual's overall health risk.